ISC FORMULA STUDENT

Season 2026/2027

Work Order – IFS09 Universal Sensor Hub

Department: Electronics & Control

Author: Andrés Sánchez, Carlota Treviño, Inés Pacheco

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1. Welcome to ISC Electronics: Sensor Acquisition & Vehicle

If you are reading this, your mission is to design the IFS09 Universal Sensor Hub. Whether it is front brake hydraulic pressure, suspension potentiometer travel, steering wheel angle, aerodynamic pitot tube airspeed, or high-accuracy chassis acceleration, this board is the universal platform that ingests, conditions, and broadcasts vehicle state across our CAN networks.

1.1 The Two Pillars: Work Order (WO) vs. Design Report (DR)

In Formula Student, no board is ever manufactured without following our two-step engineering lifecycle:

- **Work Order (WO - This Document):** Defines the 'WHAT' and the 'WHY'. It establishes the electrical requirements, pinouts, safety constraints, component benchmarks, and acceptance testing checklists before you draw a single trace in KiCad.
- **Design Report (DR - Your Deliverable):** Defines the 'HOW'. Once your KiCad schematic and PCB routing pass ERC and DRC with zero errors, you write the DR explaining your component calculations, buck regulator ripple analysis, layer stack-up, BOM, and bench validation results.

1.2 Team Dynamic & Collaborative Review

Designing an automotive-grade PCB is a collaborative process. We work in sprints, review schematics together in the workshop, and run peer reviews before ordering boards. Don't hesitate to ask questions whenever you are selecting a footprint or routing an analog trace!

2. KiCad Setup & Component Libraries

2.1 The Official Team Component Library: IFS-KicadLib

All ISC electronics designs use KiCad 8.x (or 7.x). To ensure that every manufactured board solders flawlessly at the factory, we maintain a unified component library containing validated footprints, schematic symbols, and 3D CAD models.

- **Library Setup:** Clone the official team library repository: https://github.com/isc-fs/IFS-KicadLib into your local workspace. Configure KiCad Symbol and Footprint Libraries to point to this directory.
- **Verified Automotive Parts:** Always pick parts from the IFS library first. These include our standard 0603/0805 passives, TE AMPSEAL connectors, STM32 microcontrollers, TCAN transceivers, and Bosch IMUs.

2.2 Footprint Association & 3D Clearance Verification

Every single schematic symbol must have an associated footprint with verified pin numbering. For connectors like the 23-pin TE AMPSEAL, always load the 3D STEP model in KiCad's 3D Viewer to ensure the latching mechanism clears adjacent components and enclosure walls.

3. How to Work with GitHub (HW & SW in the Same Repository)

3.1 Mandatory Action: Create Your Repository

As your very first step, create your dedicated GitHub repository under the team organization:

- **Naming Standard:** IFS09-CE-SensorHub_<YourName> (or initials).
- **Repository Template:** Initialize using the team template or mirror the structure of IFS09-TE-TeleReceiver / IFS09-CE-ControlBoard.
- **Continuous Commits:** Commit and push after every functional subcircuit (power, MCU, IMU, CAN). Never hoard uncommitted changes!

3.2 Standard Repository File Structure

- **docs/images/:** Contains exported schematic PDF/PNG previews, layer routing captures, and 3D PCB renders.
- **<Project_Name>/:** Holds all KiCad project files (.kicad_pro, .kicad_sch, .kicad_pcb).
- **SW/:** Embedded firmware, STM32CubeIDE project files, and sensor driver source code.
- **WO_IFS09_SEN_001.pdf:** Requirements and engineering specifications (this document).
- **DR_IFS09_SEN_001.pdf:** Technical defense, calculations, BOM, and bench QA sign-off.
- **Schematics-X.X.pdf:** Latest high-resolution vector PDF export of all schematic sheets.

4. How to Start in KiCad: The Schematic Editor

4.1 Essential Shortcuts & Right Sidebar Tools

- **[Esc] Pointer / Selection:** Select, move, and inspect elements.
- **[A] Add Symbol:** Open component browser to search for microcontrollers, passives, and ICs.
- **[W] Add Wire:** Draw electrical connections between symbol pins.
- **[P] Add Power / GND:** Place power rails (+24V, +12V, +5V, +3V3) and ground nodes (GND, AGND).
- **[L] Add Net Label:** Electrically connect pins across the sheet without cluttering crossing wires.
- **[Q] No-Connect Flag:** Mark intentionally unused microcontroller pins so ERC passes cleanly.

5. The Mission: IFS09 Universal Sensor Hub Architecture

You will design an automotive-grade IFS09 Universal Sensor Hub. In our race car, this board is installed in high-vibration, acoustically harsh environments (front chassis bulkhead, suspension uprights, sidepods) to read diverse sensors and stream high-rate dynamics data over dual CAN buses.

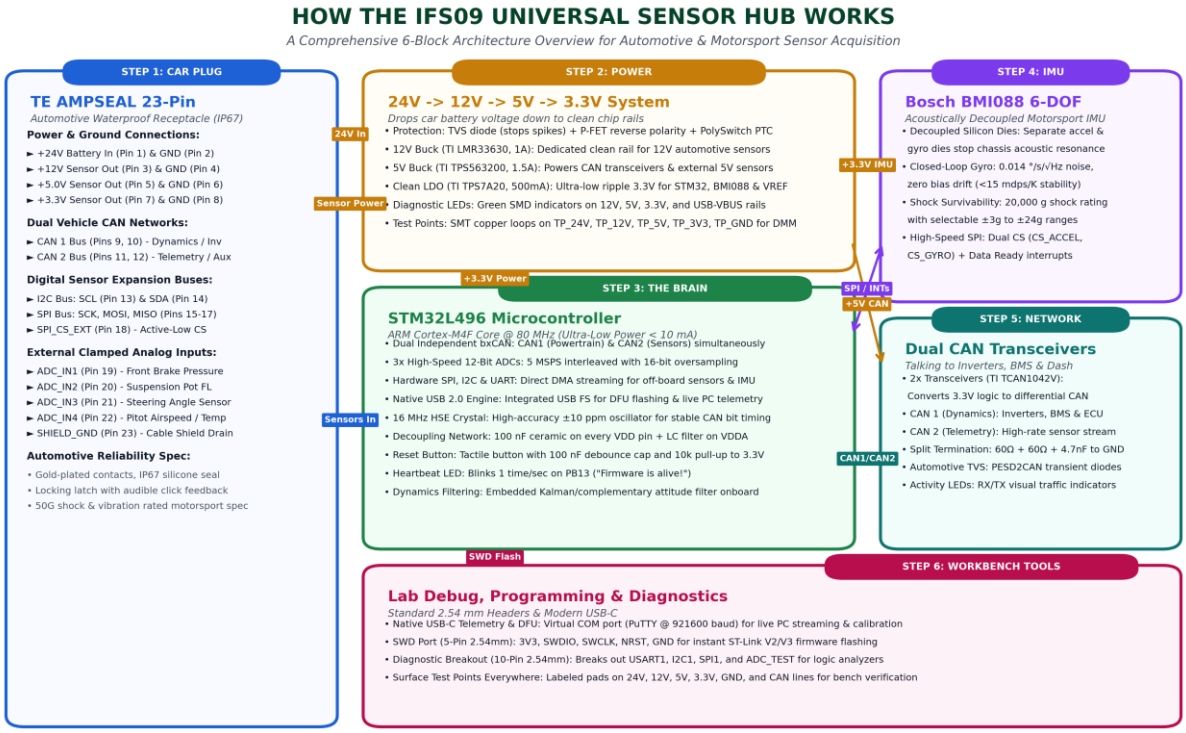


Figure 5.1: IFS09 Universal Sensor Hub Functional Block Diagram

5.1 Architecture Overview & Signal Flow

The board architecture is partitioned into six distinct, color-coded functional domains designed to ensure clean signal routing, eliminate ground loops, and isolate sensitive analog measurements from high-current switching transients:

- **Step 1: Car Plug (TE AMPSEAL 23-Pin - Royal Blue):** The sole physical interface connecting the PCB to the vehicle harness. It receives 24V battery power, exposes dual CAN buses, provides 12V, 5V, and 3.3V regulated sensor rails with isolated grounds, and routes I2C, SPI, and 4 clamped analog channels.
- **Step 2: Power System (24V -> 12V -> 5V -> 3.3V - Warm Gold):** Multi-stage automotive power conversion. Transient suppression (TVS, P-FET reverse polarity, PTC fuse) feeds a high-efficiency 12V buck (TI LMR33630) for automotive sensors, a 5V buck (TI TPS563200) for transceivers/sensors, and an ultra-low-noise 3.3V LDO (TI TPS7A20) for the MCU and IMU.
- **Step 3: The Brain (STM32L496 Microcontroller - Fresh Green):** The computational core (ARM Cortex-M4F @ 80 MHz). Operates at ultra-low active power (<10 mA), controls two independent bxCAN peripherals, samples 3x 12-bit ADCs, processes IMU kinematics, and blinks a 1 Hz heartbeat LED.

- **Step 4: Embedded High-Accuracy IMU (Bosch BMI088 - Violet):** Automotive/industrial 6-DOF IMU featuring physically decoupled gyroscope and accelerometer silicon dies, preventing chassis acoustic resonance from inducing gyro zero-rate drift.
- **Step 5: Car Network (Dual CAN Transceivers - Teal Green):** Two automotive TCAN1042V transceivers separating Vehicle Dynamics (Inverters/BMS) from Telemetry and Sensor logging, complete with 120-ohm split termination and TVS diode arrays.
- **Step 6: Workbench Tools (Lab Debug & Modern USB-C - Rose Pink):** Desktop development suite including a native USB-C port for DFU flashing and live PC telemetry (PuTTY @ 921600 baud), 5-pin SWD programming header, 10-pin diagnostic breakout, and labeled test points.

5.2 Functional Block Descriptions

- **Block 1: Car Plug (AMPSEAL 23):** Rugged IP67 sealed interface: dual power in (24V, GND), 3x regulated outputs with isolated grounds (12V, 5V, 3.3V), dual CAN channels (CAN1, CAN2), I2C, SPI, and 4 clamped analog inputs.
- **Block 2: Multi-Rail Power Plant:** Features TVS surge clamping, reverse polarity P-MOSFET, 12V buck, 5V buck, 3.3V LDO, and dedicated green status LEDs on all power rails.
- **Block 3: The Brain (STM32L496 MCU):** Ultra-low power Cortex-M4F with 16 MHz HSE crystal, 100 nF ceramic decoupling on every VDD/VDDA pin, hardware reset button, and 1 Hz Heartbeat LED on PB13.
- **Block 4: Embedded IMU (Bosch BMI088):** Decoupled 6-DOF IMU with 0.014 °/s/√Hz noise, 20,000g shock survivability, and dual high-speed SPI chip select lines.
- **Block 5: Car Network (Dual CAN):** Dual TCAN1042V transceivers with split termination and PESD2CAN TVS arrays, separating critical dynamics from sensor broadcasts.
- **Block 6: Workbench Tools (Lab Debug):** Native USB-C receptacle, standard 2.54 mm SWD header, 10-pin diagnostic breakout header, and surface test points for benchtop validation.

6. Functional Subcircuits & Design Requirements

Break your schematic into clear functional sheets. For each block, pay close attention to what is VITAL (mandatory) and the EXTRA (automotive best practices).

6.1 Power Supply Subsystem (24V -> 12V -> 5V -> 3.3V)

Qué hace: Receives 24V DC from the car's Low Voltage battery, suppresses severe inductive spikes and load dumps, and generates three regulated, protected power rails: 12V for automotive sensors, 5V for transceivers and digital sensors, and 3.3V for the MCU and IMU.

- **VITAL:** TVS diode (SMBJ33A or SMAJ28CA) on 24V input. P-MOSFET reverse polarity protection. Resettable PolySwitch PTC fuse (1.5A hold). Discrete synchronous buck converter (TI LMR33630, 36V max input, 1A output) generating 12V. Synchronous buck converter (TI TPS563200, 1.5A) generating 5.0V. Ultra-low-noise LDO (TI

TPS7A20 or MIC5365, 500mA) generating 3.3V. Bulk capacitors (10-47 µF) at the input/output of each regulator.

- **EXTRA (Why Keep the 12V Output?):** Motorsport sensors (brake line pressure transducers like Honeywell MLH, active linear suspension travel potentiometers, Texys infrared tire temperature sensors, and laser ride-height sensors) require 10V–32V excitation to deliver calibrated outputs without line drop issues. Retaining a dedicated, locally regulated 12V output is essential for Formula Student telemetry!

6.2 Microcontroller Unit (MCU — STM32L496RGT6)

Qué hace: The central computational core. Executes real-time data acquisition, samples analog sensors, manages dual independent CAN networks, processes IMU kinematics, and streams data via USB-C.

- **VITAL:** STM32L496RGT6 (LQFP64 package). External 16 MHz HSE crystal oscillator with matched 12-18 pF load capacitors. Hardware reset button with 100 nF debounce cap and 10 kΩ pull-up to 3.3V. Decoupling Capacitors (Rule #1 of Microcontrollers): Place a 100 nF ceramic capacitor on EVERY SINGLE VDD and VDDA pin as close as physically possible with direct ground vias! Green Heartbeat LED on PB13 toggling at 1 Hz.
- **EXTRA:** Dedicated LC filter (ferrite bead + 10 µF + 100 nF) on VDDA for analog ADC noise suppression. BOOT0 pull-down resistor with jumper to 3.3V for DFU bootloader flashing.

6.3 Embedded High-Accuracy IMU (Bosch BMI088)

Qué hace: Measures 3-axis acceleration and 3-axis angular velocity with high precision, low drift, and extreme vibration immunity on the racing chassis.

- **VITAL:** Bosch BMI088 LGA package. Connected via high-speed SPI to STM32L496. Separate chip select lines (CS_ACCEL, CS_GYRO) and interrupt lines (INT1, INT2, INT3) to STM32 EXTI pins. Dedicated 100 nF ceramic decoupling capacitors directly adjacent to VDD and VDDIO pins.
- **EXTRA:** Position the BMI088 near the PCB center of mass, away from high-current switching inductors and mounting holes. Unbroken ground plane beneath the IMU footprint.

6.4 Dual CAN Bus Interfaces (CAN1 & CAN2)

Qué hace: Connects the sensor hub to the vehicle's high-speed differential CAN networks (Inverters, BMS, Dash, and Telemetry).

- **VITAL:** 2x automotive CAN transceivers (TI TCAN1042V-Q1) powered from 5V with 3.3V logic compatibility. 120-ohm bus termination on each channel selectable via 2-pin SMD jumper. 100 nF decoupling capacitor across VCC/GND on each transceiver.
- **EXTRA:** Split termination network (two 60-ohm 1% resistors in series with a 4.7 nF capacitor to GND) for superior common-mode noise suppression. Automotive TVS diode arrays (PESD2CAN or NUP2105L) on CAN_H and CAN_L lines.

6.5 Automotive Connector: TE AMPSEAL 23-Pin Header

Qué hace: Rugged, waterproof physical interface connecting the PCB to the vehicle wiring harness. Below is the complete pinout mapping:

Pin #	Signal Name	Voltage / Rating	What it connects to
1	+24V_IN	9V – 36V DC, Max 1.5A	Main Vehicle Low Voltage (LV) Bus Power Input
2	GND_IN	0V Reference	Main Power Return Ground from LV Battery
3	+12V_OUT	12.0V ± 2%, Max 800 mA	Regulated 12V Power Output for Automotive Transducers
4	GND_12V	0V Clean Ground	Dedicated Ground Return for 12V Sensor Circuitry
5	+5V_OUT	5.0V ± 2%, Max 1.0A	Regulated 5V Power Output for Standard Sensors & Encoders
6	GND_5V	0V Clean Ground	Dedicated Ground Return for 5V Sensor Circuitry
7	+3.3V_OUT	3.3V ± 1%, Max 300 mA	Regulated Low-Noise 3.3V Output for Digital Sensors
8	GND_3V3	0V Clean Ground	Dedicated Ground Return for 3.3V Digital Circuitry
9	CAN1_H	ISO 11898-2 Standard	CAN Bus 1 High (Vehicle Dynamics & Powertrain)
10	CAN1_L	ISO 11898-2 Standard	CAN Bus 1 Low (Vehicle Dynamics & Powertrain)
11	CAN2_H	ISO 11898-2 Standard	CAN Bus 2 High (Telemetry & Secondary Sensors)
12	CAN2_L	ISO 11898-2 Standard	CAN Bus 2 Low (Telemetry & Secondary Sensors)
13	I2C_SCL	3.3V Open-Drain, 4.7kΩ	I2C Serial Clock Line for External Sensor Bus
14	I2C_SDA	3.3V Open-Drain, 4.7kΩ	I2C Serial Data Line for External Sensor Bus
15	SPI_SCK	3.3V CMOS, Max 10 MHz	Master SPI Clock Output to Off-Board Peripherals
16	SPI_MOSI	3.3V CMOS Logic	Master SPI Data Output (MOSI)
17	SPI_MISO	3.3V CMOS Logic	Master SPI Data Input (MISO)
18	SPI_CS_EXT	3.3V Active-Low	Dedicated Chip Select for External Off-Board SPI Module
19	ADC_IN1	0.0V – 3.3V Clamped	Analog Channel 1 (Front Brake Line Pressure Sensor)
20	ADC_IN2	0.0V – 3.3V Clamped	Analog Channel 2 (Suspension Travel Linear Potentiometer)
21	ADC_IN3	0.0V – 3.3V Clamped	Analog Channel 3 (Steering Wheel Angle Hall Sensor)
22	ADC_IN4	0.0V – 3.3V Clamped	Analog Channel 4 (Dynamic Pitot Airspeed / Tire Temp IR)
23	SHIELD_GND	Chassis Ground	Cable Harness Braided Shield Drain Connection

6.6 Analog Conditioning & Sensor Front-End (ADC1–ADC4)

Qué hace: Conditions noisy external analog sensor signals and protects microcontroller ADC pins against over-voltage and negative transient spikes.

- **VITAL:** Passive first-order RC anti-aliasing filter (100 Ω resistor + 10 nF C0G capacitor, f_cutoff ≈ 159 kHz) on each analog channel. Dual Schottky clamp diodes (BAT54S) tied between AGND, the ADC pin, and 3.3V to clamp voltage excursions.
- **EXTRA:** Precision 0.1% resistor dividers for scaling 0-5V or 0-10V automotive sensors down to the 0-3.3V ADC range.

6.7 Expansion & Debug Headers (2.54 mm Pitch) & Native USB-C

Qué hace: Allows bench programming, calibration, logic analyzer inspection, and serial telemetry without crimping AMPSEAL pins.

- **VITAL:** SWD Programming Header (5-pin: 3V3, SWDIO, SWCLK, NRST, GND) for ST-Link flashing. Native USB-C receptacle wired to STM32 USB pins with 5.1 kΩ pull-down resistors on CC1/CC2 and USBLC6-2SC6 ESD protection.
- **EXTRA:** MBR0520 Schottky barrier diode isolating vehicle 24V power from host computer USB VBUS. 10-pin diagnostic breakout header breaking out USART1, I2C1, SPI1, and ADC_TEST.

6.8 Diagnostic Status LEDs & Test Points

Qué hace: Allows anyone in the workshop or trackside pits to instantly diagnose hardware health without an oscilloscope.

- **Diagnostic LEDs:** Green SMD LEDs on 12V, 5V, 3.3V, and USB rails. Green MCU Heartbeat LED on PB13 (blinks once per second proving firmware is alive). CAN activity LEDs on transceiver TX/RX lines.
- **Test Points Everywhere:** Labeled test pads (min 1.0 mm pad diameter) for TP_24V, TP_12V, TP_5V, TP_3V3, TP_GND, TP_CAN1_H, TP_CAN1_L, TP_NRST, TP_SWDIO, TP_SWCLK.

7. Acceptance Testing Checklist (Post-Assembly Bench QA)

When your manufactured board arrives from the assembly factory, follow this step-by-step verification checklist:

- **Step 1 (Passive Short-Circuit Check):** Measure resistance across supply test points to GND with a multimeter. All readings must be > 10 kΩ on 12V/5V and > 1 kΩ on 3.3V before power-up (no solder bridges).
- **Step 2 (Power-Up & Regulation):** Connect 24V bench power supply (current-limited to 200 mA). Verify V(TP_12V) = 12.0V ± 0.3V, V(TP_5V) = 5.0V ± 0.15V, V(TP_3V3) = 3.3V ± 0.05V, and verify that all power LEDs illuminate.
- **Step 3 (SWD Detection & MCU Flash):** Connect ST-Link to SWD header. Detect target in STM32CubeProgrammer and flash test firmware to blink the Heartbeat LED (PB13).
- **Step 4 (Bosch BMI088 SPI Check):** Read WHO_AM_I registers over SPI @ 10 MHz. Accelerometer returns 0x1E, Gyroscope returns 0x0F. Verify stationary 1g vertical acceleration.

- **Step 5 (Dual CAN Loopback):** Connect CAN1 & CAN2 to a USB-CAN adapter (PCAN / Vector) and verify transmitting and receiving packets at 1 Mbps with 0 dropped frames.
- **Step 6 (ADC Accuracy Test):** Apply 0.500V, 1.500V, 2.500V to ADC inputs and verify ADC readings within ±1% precision.
- **Step 7 (USB-C Telemetry Stream):** Connect USB-C to PC. Open terminal at 921600 baud. Stream continuous multi-sensor telemetry for 30 minutes with zero drops.

8. Quality Assurance Bench Sign-Off Matrix

This formal sign-off matrix must be completed, dated, and signed by the electronics assembly inspector before the Sensor Hub is installed in the vehicle:

Test Parameter	Specified Acceptance Target	Measured Value	Pass / Fail	Inspector / Date
1. Unpowered Impedance	R > 10 kΩ (12V/5V), > 1 kΩ (3.3V) to GND			
2. 12V Regulated Sensor Rail	12.0 V ± 0.3 V (Ripple < 50 mVp-p)			
3. 5V Regulated Sensor Rail	5.00 V ± 0.15 V (Ripple < 30 mVp-p)			
4. 3.3V Regulated Sensor Rail	3.30 V ± 0.05 V (Ripple < 15 mVp-p)			
5. MCU Heartbeat LED	1.0 Hz Toggle, SWD Programming OK			
6. Bosch BMI088 SPI ID Check	Accel ID: 0x1E, Gyro ID: 0x0F			
7. Dual CAN 1 & 2 Bus (1 Mbps)	0% packet drop / 100k test frames			
8. 4-Ch ADC Linearity & Noise	INL < 2 LSB, RMS Noise < 1.5 mV			

9. What You Need to Submit: The Design Report

Once your schematic and PCB routing pass KiCad DRC and ERC with 0 errors, submit a formal Design Report containing:

- **1. Executive Summary:** Brief functional overview of your board and its specific sensor role on the car.
- **2. Component Calculations & Justifications:** Buck converter component selection (inductor saturation current, diode rating, ripple calculation), LDO dropout, and decoupling capacitor network.
- **3. Schematic Export:** Clean, high-resolution vector PDF schematic export (Schematics-1.0.pdf).
- **4. PCB Layout Showcase:** 2D layout captures showing the ground plane and 3D renders showing mechanical clearance for the TE AMPSEAL connector and M3 mounting holes (< 80x80 mm).
- **5. Bill of Materials (BOM):** Complete spreadsheet with manufacturer part numbers (MPN), component packages, and supplier sourcing links.